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THE IMPACT OF PHYSICAL ACTIVITY ON SURVIVAL OUTCOMES,  
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WITH BREAST, COLORECTAL, AND PROSTATE CANCER: A  
NARRATIVE REVIEW

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# THE IMPACT OF PHYSICAL ACTIVITY ON SURVIVAL OUTCOMES, TREATMENT RESPONSE, AND QUALITY OF LIFE IN PATIENTS WITH BREAST, COLORECTAL, AND PROSTATE CANCER: A NARRATIVE REVIEW

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## ABSTRACT

Physical activity is increasingly recognized as an important component of comprehensive cancer care. This narrative review summarizes current evidence on the effects of exercise on survival outcomes, treatment response, and quality of life in patients with breast, colorectal, and prostate cancer. Regular exercise improves physical function, reduces cancer-related fatigue, enhances treatment tolerance, and may improve survival, particularly in breast and colorectal cancer. These benefits are mediated through anti-inflammatory, metabolic, and immunological mechanisms. Individualized exercise programmes should be integrated into routine oncology care.

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## KEYWORDS

Physical Activity, Exercise Oncology, Breast Cancer, Colorectal Cancer, Prostate Cancer

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## Introduction

Cancer remains one of the leading causes of morbidity and mortality worldwide, accounting for nearly 20 million new cases and approximately 9.7 million deaths annually. Breast, colorectal, and prostate cancers are among the most frequently diagnosed malignancies and collectively represent a substantial proportion of the global cancer burden. Although advances in early detection, surgery, radiotherapy, systemic therapies, and supportive care have significantly improved survival rates, the growing number of cancer survivors has shifted attention toward strategies aimed at improving long-term outcomes, reducing treatment-related toxicities, and enhancing quality of life.

Physical activity has emerged as one of the most promising non-pharmacological interventions in modern oncology. Over the past two decades, accumulating evidence from epidemiological studies, randomized controlled trials, and systematic reviews has demonstrated that regular exercise is associated with numerous physiological and psychological benefits in patients with cancer. These include improvements in cardiorespiratory fitness, preservation of skeletal muscle mass, better metabolic control, reduced systemic inflammation, enhanced immune function, decreased cancer-related fatigue, and improved health-related quality of life.

The biological mechanisms through which exercise may influence cancer progression are complex and multifactorial. Exercise has been shown to modulate chronic inflammation, improve insulin sensitivity, reduce circulating insulin-like growth factor-1 (IGF-1), stimulate the release of antitumor myokines such as irisin, and enhance natural killer (NK) cell activity and immune surveillance. Collectively, these mechanisms may contribute to slowing tumour progression, improving treatment efficacy, and reducing the risk of recurrence.

Among solid malignancies, breast, colorectal, and prostate cancers have been the focus of the largest body of research investigating exercise oncology. In breast cancer, regular physical activity has consistently been associated with lower all-cause and breast cancer-specific mortality, improved treatment tolerance, and better physical and psychological functioning. Similarly, colorectal cancer survivors who engage in regular exercise demonstrate improved functional capacity, reduced postoperative complications, enhanced recovery, and potentially improved survival outcomes. In men with prostate cancer, particularly those receiving androgen deprivation therapy, structured exercise programmes have been shown to preserve bone and muscle mass, reduce fatigue, improve physical performance, and mitigate treatment-related adverse effects.

Despite the rapidly expanding evidence base, important uncertainties remain regarding the optimal exercise prescription, including exercise modality, intensity, frequency, and duration across different cancer types and treatment phases. Furthermore, although observational studies consistently report associations between higher levels of physical activity and improved survival, randomized clinical trials evaluating long-term oncological endpoints remain relatively limited. Consequently, further research is required to establish individualized exercise prescriptions that maximize therapeutic benefits while ensuring patient safety.

Recent international recommendations increasingly recognize physical activity as an essential component of comprehensive cancer care. Current oncology guidelines encourage healthcare professionals to prescribe individualized exercise programmes whenever clinically appropriate, emphasizing that physical activity should be integrated throughout the entire cancer continuum, from diagnosis to survivorship and palliative care.

The aim of this narrative review is to summarize current evidence regarding the impact of physical activity on survival outcomes, treatment response, and health-related quality of life in patients with breast, colorectal, and prostate cancer. Particular emphasis is placed on the biological mechanisms underlying exercise-induced benefits, findings from recent systematic reviews and meta-analyses, current clinical recommendations, and future directions in exercise oncology.

## Methods

This narrative review was conducted to summarize current evidence regarding the effects of physical activity on survival outcomes, treatment response, and health-related quality of life in patients with breast, colorectal, and prostate cancer. Relevant peer-reviewed articles were identified through a literature search of major biomedical databases, including PubMed, Scopus, and Web of Science. The review focused primarily on systematic reviews, meta-analyses, randomized controlled trials, prospective cohort studies, and current international clinical guidelines published within the last decade, while landmark earlier studies were included where appropriate.

Studies investigating exercise interventions performed before, during, or after cancer treatment were considered eligible if they evaluated at least one clinically relevant outcome, including overall survival, cancer-specific survival, disease recurrence, treatment-related adverse events, physical function, cancer-related fatigue, or quality of life. Preference was given to high-quality evidence and studies involving patients diagnosed with breast, colorectal, or prostate cancer.

Due to the narrative nature of this review, a formal systematic review methodology and quantitative meta-analysis were not performed. Instead, the available evidence was critically synthesized to provide a comprehensive overview of current knowledge and to identify important areas requiring further investigation.

### **Biological Mechanisms Linking Physical Activity with Cancer Outcomes**

The beneficial effects of physical activity in oncology extend beyond improvements in physical fitness and functional capacity. Increasing evidence suggests that exercise induces a wide range of systemic physiological adaptations capable of influencing tumour biology, host metabolism, immune function, and the tumour microenvironment. These mechanisms may contribute to improved treatment tolerance, reduced disease progression, and enhanced long-term survival in patients with cancer

One of the most extensively investigated mechanisms is the anti-inflammatory effect of regular exercise. Chronic low-grade inflammation plays an important role in carcinogenesis and tumour progression by promoting angiogenesis, DNA damage, cellular proliferation, and immune evasion. Exercise has been shown to reduce circulating concentrations of pro-inflammatory cytokines, including interleukin-6 (IL-6), tumour necrosis factor- $\alpha$  (TNF- $\alpha$ ), and C-reactive protein (CRP), while simultaneously increasing anti-inflammatory mediators. This shift toward a less inflammatory systemic environment may inhibit tumour growth and improve responses to anticancer therapies

Physical activity also exerts substantial metabolic effects. Obesity, insulin resistance, and hyperinsulinemia have been associated with an increased risk of several malignancies, particularly breast and colorectal cancer. Regular exercise improves insulin sensitivity, lowers circulating insulin concentrations, and reduces insulin-like growth factor-1 (IGF-1) signalling, thereby limiting activation of proliferative pathways such as PI3K/Akt/mTOR. These metabolic adaptations may reduce tumour cell proliferation while increasing susceptibility to apoptosis

Another important mechanism involves modulation of the immune system. Acute bouts of exercise stimulate the mobilisation of natural killer (NK) cells, cytotoxic T lymphocytes, dendritic cells, and other immune cell populations involved in tumour surveillance. Repeated exercise sessions appear to enhance immune competence and improve recognition and elimination of malignant cells. Experimental studies additionally suggest that exercise-induced increases in circulating catecholamines facilitate NK-cell redistribution toward tumour tissue, potentially contributing to reduced tumour growth

Skeletal muscle has recently been recognised as an endocrine organ capable of producing biologically active molecules known as myokines. During muscular contraction, cytokines and peptides such as irisin, interleukin-15, secreted protein acidic and rich in cysteine (SPARC), and brain-derived neurotrophic factor (BDNF) are released into the circulation. Several of these myokines demonstrate anti-proliferative, anti-inflammatory, and anti-metastatic properties in experimental cancer models, suggesting that skeletal muscle actively participates in the regulation of tumour biology rather than functioning solely as a target of cancer-associated cachexia

Exercise may also improve the tumour microenvironment by enhancing vascular function and tissue perfusion. Increased blood flow facilitates oxygen delivery, reducing tumour hypoxia—a condition associated with resistance to chemotherapy and radiotherapy. Improved vascularisation may therefore increase the effectiveness of systemic anticancer treatment while reducing treatment-related toxicity to healthy tissues .

Collectively, these biological adaptations provide a mechanistic rationale for the growing clinical evidence supporting physical activity as an important component of comprehensive cancer care. Although many molecular pathways remain incompletely understood, current evidence indicates that exercise influences multiple hallmarks of cancer simultaneously, highlighting its potential as an effective adjunctive therapy throughout the cancer continuum.

## Physical Activity in Breast Cancer

Breast cancer is the most frequently diagnosed malignancy among women worldwide and remains a leading cause of cancer-related mortality despite substantial improvements in screening, diagnosis, and treatment. Advances in surgery, systemic therapies, radiotherapy, endocrine therapy, and targeted treatment have significantly increased survival rates, shifting attention toward interventions capable of improving long-term survivorship and reducing treatment-related morbidity .

Among lifestyle interventions, physical activity has consistently demonstrated beneficial effects throughout the entire breast cancer continuum, including the pre-treatment, active treatment, and survivorship phases. Evidence from large prospective cohort studies suggests that women engaging in regular moderate-to-vigorous physical activity after diagnosis experience significantly lower all-cause and breast cancer-specific mortality compared with physically inactive patients. Meta-analyses have estimated reductions in overall mortality ranging from approximately 20% to 40%, although the magnitude of benefit varies depending on exercise intensity, duration, tumour characteristics, and patient adherence.

Exercise also plays an important role during active cancer treatment. Chemotherapy, endocrine therapy, and radiotherapy are frequently associated with fatigue, reduced cardiorespiratory fitness, muscle wasting, weight gain, psychological distress, and impaired physical functioning. Randomized controlled trials consistently demonstrate that structured aerobic, resistance, or combined exercise programmes significantly attenuate these adverse effects. Patients participating in supervised exercise interventions report lower fatigue scores, greater muscle strength, improved aerobic capacity, and enhanced physical functioning compared with those receiving standard care .

Cancer-related fatigue deserves particular attention, as it represents one of the most common and distressing symptoms experienced by breast cancer patients. Unlike ordinary fatigue, cancer-related fatigue is persistent, disproportionate to activity level, and often poorly relieved by rest. Exercise is currently regarded as one of the most effective non-pharmacological interventions for managing this symptom. Both aerobic exercise and resistance training have demonstrated clinically meaningful reductions in fatigue severity while simultaneously improving sleep quality, emotional well-being, and overall quality of life .

Emerging evidence further suggests that exercise may influence treatment efficacy through biological mechanisms affecting tumour behaviour and host physiology. Regular physical activity improves insulin sensitivity, reduces adiposity, decreases circulating inflammatory mediators, and enhances immune surveillance. These physiological adaptations may create a systemic environment less favourable for tumour progression while improving tolerance to systemic anticancer therapies. Experimental studies additionally indicate that exercise-induced myokines and increased natural killer cell activity may contribute to inhibition of tumour growth, although the clinical significance of these mechanisms continues to be investigated .

Exercise is also particularly valuable for women receiving endocrine therapy, especially aromatase inhibitors. Musculoskeletal pain, reduced bone mineral density, weight gain, and declining physical function frequently compromise long-term treatment adherence. Several randomized trials have shown that regular resistance training combined with weight-bearing aerobic exercise improves muscular strength, preserves bone health, reduces treatment-related joint symptoms, and enhances overall physical performance. Consequently, exercise interventions may indirectly improve adherence to endocrine therapy by reducing treatment-associated adverse effects .

Beyond physical outcomes, psychological benefits represent another important aspect of exercise oncology. Breast cancer survivors frequently experience anxiety, depression, fear of recurrence, body image disturbances, and reduced self-esteem. Exercise programmes, particularly those performed in supervised group settings, have been associated with significant improvements in emotional functioning, social participation, self-efficacy, and overall health-related quality of life. These benefits appear to persist long after completion of structured interventions, supporting the incorporation of long-term physical activity into survivorship care plans .

Current international guidelines recommend that breast cancer survivors perform at least 150–300 minutes of moderate-intensity aerobic exercise or 75–150 minutes of vigorous-intensity activity weekly, together with resistance training involving major muscle groups at least twice per week. Nevertheless, exercise prescriptions should always be individualized according to treatment stage, functional capacity, comorbidities, and patient preferences to maximize adherence and clinical benefit .

Overall, the available evidence strongly supports physical activity as an integral component of modern breast cancer care. Regular exercise not only improves physical function and quality of life but may also contribute to improved long-term survival and enhanced tolerance to oncological treatment. Future research should focus on identifying optimal exercise prescriptions for specific molecular subtypes of breast cancer and determining whether individualized exercise interventions can further improve clinical outcomes.

## Physical Activity in Colorectal Cancer

Colorectal cancer (CRC) is the third most commonly diagnosed malignancy worldwide and remains one of the leading causes of cancer-related mortality. Despite considerable improvements in screening programmes, surgical techniques, chemotherapy, targeted therapies, and multidisciplinary management, many patients continue to experience substantial treatment-related morbidity and long-term functional impairment. Consequently, increasing attention has been directed toward supportive interventions capable of improving both oncological outcomes and survivorship. Among these, physical activity has emerged as one of the most promising non-pharmacological strategies .

Extensive epidemiological evidence indicates that physically active individuals have a lower risk of developing colorectal cancer than sedentary populations. Furthermore, increasing evidence suggests that maintaining or adopting regular physical activity following diagnosis is associated with improved overall survival, colorectal cancer-specific survival, and reduced disease recurrence. Large prospective cohort studies have demonstrated that patients performing moderate-to-vigorous physical activity after diagnosis experience significantly lower mortality compared with inactive individuals, even after adjustment for age, body mass index, tumour stage, and treatment characteristics .

The survival benefits associated with exercise appear to be mediated through multiple biological pathways. Colorectal carcinogenesis is strongly influenced by obesity, insulin resistance, chronic systemic inflammation, and alterations in metabolic signalling. Regular physical activity improves insulin sensitivity, decreases circulating insulin and insulin-like growth factor-1 (IGF-1), reduces visceral adiposity, and suppresses chronic inflammatory processes. These adaptations may inhibit tumour cell proliferation, angiogenesis, and metastatic potential while promoting apoptosis and improving host immune surveillance .

Exercise also plays an important role during active treatment for colorectal cancer. Surgical resection, chemotherapy, and radiotherapy frequently lead to reductions in muscle mass, impaired aerobic capacity, persistent fatigue, gastrointestinal symptoms, and decreased physical function. Several randomized controlled trials have demonstrated that structured exercise programmes initiated before or during treatment improve cardiorespiratory fitness, preserve lean body mass, accelerate postoperative recovery, and facilitate earlier return to normal daily activities .

Prehabilitation has recently become an important component of perioperative colorectal cancer care. This approach involves implementing exercise training before surgery with the aim of improving physiological reserve and reducing postoperative complications. Evidence suggests that multimodal prehabilitation programmes combining aerobic exercise, resistance training, nutritional optimization, and psychological support improve functional capacity prior to surgery and may reduce hospital length of stay while enhancing postoperative recovery. Although additional large randomized trials are needed, current findings support the integration of prehabilitation into selected surgical pathways .

Cancer-related fatigue remains another major challenge for colorectal cancer survivors. Similar to findings reported in breast cancer, exercise interventions have consistently demonstrated reductions in fatigue severity together with improvements in muscular strength, physical functioning, and health-related quality of life. Aerobic exercise, resistance training, and combined exercise programmes all appear effective, although combined interventions may provide the greatest overall benefit due to simultaneous improvements in cardiovascular fitness and muscle strength .

The psychological benefits of physical activity are equally important. Colorectal cancer survivors frequently experience anxiety, depression, fear of recurrence, social isolation, and reduced confidence following treatment, particularly among patients living with temporary or permanent stomas. Participation in supervised exercise programmes has been associated with improvements in emotional well-being, self-efficacy, body image, and social functioning. Group-based exercise interventions may additionally provide valuable peer support, contributing to long-term adherence and improved psychological adaptation following cancer treatment .

Recent investigations have also explored the interaction between exercise and the tumour immune microenvironment. Experimental and translational studies suggest that regular physical activity enhances natural killer cell activity, improves T-cell function, and modulates cytokine profiles, thereby potentially increasing responsiveness to systemic therapies, including immunotherapy. Although clinical evidence remains limited, these findings represent an important area of ongoing research and may further strengthen the role of exercise as an adjunctive component of colorectal cancer treatment .

Current international recommendations encourage colorectal cancer survivors to accumulate at least 150 minutes of moderate-intensity aerobic exercise per week while incorporating resistance training on two or

more days weekly. However, exercise programmes should be individualized according to disease stage, treatment modality, postoperative recovery, comorbidities, and baseline physical fitness. Even modest increases in daily physical activity appear to confer clinically meaningful health benefits, emphasizing that some activity is preferable to complete inactivity .

Overall, the available literature consistently supports physical activity as an important component of comprehensive colorectal cancer care. Exercise not only improves physical performance and quality of life but is also associated with favourable survival outcomes and enhanced recovery throughout the cancer continuum. Future research should focus on determining the optimal exercise prescription for different molecular subtypes of colorectal cancer, identifying biomarkers predictive of exercise responsiveness, and clarifying the interaction between physical activity and modern systemic therapies, including immunotherapy

### **Physical Activity in Prostate Cancer**

Prostate cancer is the second most frequently diagnosed cancer among men worldwide and remains one of the leading causes of cancer-related mortality. Improvements in prostate-specific antigen (PSA) screening, diagnostic imaging, surgical techniques, radiotherapy, androgen deprivation therapy (ADT), and novel systemic treatments have substantially increased survival, particularly among patients diagnosed with localized disease. As survival has improved, greater emphasis has been placed on interventions that reduce treatment-related adverse effects and improve long-term health outcomes. Physical activity has emerged as one of the most effective supportive strategies throughout the prostate cancer continuum .

Unlike breast and colorectal cancer, where exercise has primarily been investigated for its association with survival outcomes, research in prostate cancer has largely focused on mitigating the adverse effects of ADT. Although ADT remains a cornerstone of treatment for advanced and high-risk disease, it is associated with numerous metabolic and musculoskeletal complications, including loss of lean muscle mass, increased fat mass, reduced bone mineral density, insulin resistance, cardiovascular dysfunction, fatigue, and diminished physical performance. These treatment-related effects substantially impair patients' quality of life and increase the risk of long-term morbidity .

A growing number of randomized controlled trials have demonstrated that structured exercise programmes effectively counteract many of these adverse effects. Resistance training, either alone or combined with aerobic exercise, has consistently been shown to preserve skeletal muscle mass, improve muscular strength, maintain bone health, and enhance functional mobility in men receiving ADT. Improvements in handgrip strength, lower-limb muscle function, balance, and walking capacity have been observed following supervised exercise interventions lasting between 12 weeks and 12 months .

Aerobic exercise also provides important cardiovascular and metabolic benefits. Men undergoing ADT are at increased risk of obesity, dyslipidaemia, hypertension, type 2 diabetes mellitus, and cardiovascular disease. Regular moderate-to-vigorous physical activity improves cardiorespiratory fitness, reduces visceral adiposity, enhances insulin sensitivity, and contributes to better blood pressure and lipid control. These benefits are particularly relevant given that cardiovascular disease represents one of the leading non-cancer causes of death among prostate cancer survivors .

Cancer-related fatigue is another common consequence of prostate cancer treatment and significantly limits patients' daily functioning. Multiple systematic reviews have identified exercise as one of the most effective interventions for reducing fatigue while simultaneously improving physical endurance and overall functional capacity. Combined aerobic and resistance exercise programmes appear to produce the greatest improvements, particularly when initiated early during systemic treatment .

Exercise additionally exerts important psychological benefits. Men diagnosed with prostate cancer frequently report anxiety, depressive symptoms, fear of disease progression, reduced self-confidence, and impaired sexual health following treatment. Supervised exercise programmes have been associated with significant improvements in emotional well-being, self-esteem, cognitive function, and health-related quality of life. Participation in group-based exercise may also enhance social support and improve long-term adherence to healthy lifestyle behaviours .

Although the survival benefits of physical activity in prostate cancer are less extensively documented than in breast or colorectal cancer, observational studies suggest that physically active men experience lower all-cause mortality and may have a reduced risk of prostate cancer-specific mortality. Higher levels of recreational physical activity performed after diagnosis have been associated with improved long-term outcomes, although the strength of this association varies across studies. Potential biological mechanisms

include reductions in systemic inflammation, improvements in metabolic regulation, enhanced immune surveillance, and modulation of androgen signalling pathways .

Recent translational research has further highlighted the influence of exercise on tumour biology. Physical activity may alter the tumour microenvironment by improving tissue perfusion, reducing chronic inflammation, increasing antioxidant capacity, and stimulating the secretion of exercise-induced myokines with potential antitumour activity. Furthermore, improvements in body composition resulting from regular exercise may reduce obesity-related hormonal disturbances that have been implicated in prostate cancer progression. Nevertheless, additional prospective studies are required to clarify whether these biological adaptations directly translate into improved oncological outcomes .

Current international guidelines recommend that prostate cancer survivors perform at least 150 minutes of moderate-intensity aerobic exercise each week in combination with progressive resistance training involving all major muscle groups at least twice weekly. Flexibility and balance exercises are also encouraged, particularly among older adults receiving ADT, to reduce the risk of falls and preserve functional independence. Exercise programmes should be individualized according to disease stage, treatment modality, comorbidities, and baseline physical fitness while taking patient preferences into account to maximize long-term adherence .

Overall, the available evidence strongly supports the integration of structured exercise into routine prostate cancer management. Physical activity effectively mitigates many adverse effects of treatment, improves physical function and quality of life, and may contribute to favourable long-term clinical outcomes. Future research should focus on identifying optimal exercise prescriptions during different stages of prostate cancer treatment and further elucidating the molecular mechanisms through which exercise may influence tumour progression and survival .

## Discussion

This narrative review summarizes current evidence regarding the impact of physical activity on survival outcomes, treatment response, and health-related quality of life in patients with breast, colorectal, and prostate cancer. Overall, the available literature consistently supports the incorporation of structured exercise into comprehensive cancer care. Although the strength of evidence varies across cancer types and clinical outcomes, the collective findings indicate that regular physical activity provides substantial physiological, psychological, and potentially oncological benefits.

One of the most consistent observations across all three malignancies is the positive effect of exercise on health-related quality of life. Regardless of tumour type or treatment modality, physically active patients generally report lower levels of cancer-related fatigue, better physical functioning, improved emotional well-being, and greater independence in daily activities. These findings have been confirmed by numerous randomized controlled trials and systematic reviews, making exercise one of the few supportive interventions supported by high-quality evidence for improving multiple patient-reported outcomes simultaneously. The consistency of these findings across diverse patient populations strengthens the recommendation that physical activity should become a routine component of survivorship care .

The evidence regarding survival outcomes is strongest for breast and colorectal cancer. Large prospective cohort studies consistently demonstrate that patients who maintain regular moderate-to-vigorous physical activity after diagnosis experience lower all-cause and cancer-specific mortality than sedentary individuals. While the exact magnitude of this benefit differs among studies, the overall direction of the association remains remarkably consistent. Nevertheless, most available evidence originates from observational research, limiting the ability to establish causality. Physically active individuals frequently differ from inactive patients with respect to diet, smoking habits, body composition, socioeconomic status, and healthcare utilization, all of which may contribute to improved outcomes despite statistical adjustment for known confounders .

In contrast, evidence linking exercise to improved survival in prostate cancer remains less robust. Most randomized trials in prostate cancer have focused on functional outcomes rather than oncological endpoints because prolonged follow-up is required to detect differences in survival. Consequently, although observational studies suggest a favourable association between physical activity and reduced mortality, additional long-term randomized studies are necessary before definitive conclusions can be drawn. Future research should determine whether improvements in metabolic health and immune function ultimately translate into measurable reductions in disease progression and cancer-specific mortality .

The biological mechanisms underlying exercise-induced benefits continue to be actively investigated. Increasing evidence suggests that exercise simultaneously influences several hallmarks of cancer through modulation of chronic inflammation, insulin signalling, adipokine secretion, immune surveillance, oxidative stress, and skeletal muscle-derived myokines. Rather than acting through a single pathway, physical activity appears to induce multiple complementary physiological adaptations that collectively create a systemic environment less favourable for tumour progression. This multifactorial mechanism may explain why exercise demonstrates beneficial effects across different cancer types despite substantial differences in tumour biology .

Another important finding emerging from the literature is that exercise appears to improve tolerance to anticancer treatment. Patients participating in structured exercise programmes frequently demonstrate better cardiorespiratory fitness, greater muscle strength, reduced treatment-related fatigue, and improved physical performance throughout chemotherapy, radiotherapy, endocrine therapy, and androgen deprivation therapy. Improved functional capacity may allow patients to maintain planned treatment intensity while reducing interruptions caused by adverse effects. Although direct evidence demonstrating improved treatment efficacy remains limited, better treatment adherence may indirectly contribute to improved long-term outcomes .

Despite encouraging findings, several important limitations should be acknowledged. Considerable heterogeneity exists among published studies regarding exercise modality, frequency, intensity, duration, supervision, patient characteristics, treatment stage, and measured outcomes. Such methodological variability complicates direct comparisons between studies and limits the development of universally applicable exercise prescriptions. Furthermore, adherence to prescribed exercise interventions varies substantially, particularly among older adults and patients undergoing intensive systemic treatment. Future randomized trials should adopt standardized exercise protocols and consistently report intervention characteristics to facilitate comparison between studies and strengthen future evidence synthesis .

Another limitation concerns the underrepresentation of specific patient populations. Most available studies include patients with good performance status and relatively few comorbidities, whereas evidence remains limited for frail older adults, individuals with metastatic disease, and patients receiving novel immunotherapeutic or targeted treatments. Additionally, relatively few studies evaluate long-term adherence to physical activity following completion of supervised exercise programmes, despite the importance of sustained lifestyle modification for survivorship care.

From a clinical perspective, the available evidence strongly supports integrating exercise professionals into multidisciplinary oncology teams. Individualized exercise prescriptions developed through collaboration between oncologists, physiotherapists, exercise physiologists, rehabilitation specialists, and oncology nurses may maximize patient safety while optimizing therapeutic benefits. Routine assessment of physical activity should therefore become an integral component of oncology consultations, allowing clinicians to identify inactive patients who may benefit most from structured exercise interventions .

Future research should move beyond simply determining whether exercise is beneficial and instead focus on identifying the optimal exercise prescription for specific patient populations. Precision exercise oncology, in which exercise programmes are tailored according to tumour biology, treatment modality, molecular biomarkers, genetic background, and individual physiological characteristics, represents a promising direction for future investigation. Furthermore, integrating wearable technologies, digital health platforms, and artificial intelligence may improve long-term monitoring, adherence, and personalization of exercise interventions.

Overall, the current evidence indicates that physical activity should no longer be regarded solely as a lifestyle recommendation but rather as an important component of comprehensive cancer management. Although further high-quality randomized trials remain necessary to clarify several unresolved questions, the consistent improvements observed in physical function, treatment tolerance, and quality of life strongly support the routine incorporation of structured exercise into modern oncology practice.

## Conclusions

The available evidence demonstrates that physical activity plays an important role in the comprehensive management of patients with breast, colorectal, and prostate cancer. Across all three malignancies, regular exercise has consistently been associated with improvements in physical function, cardiorespiratory fitness, muscular strength, treatment tolerance, and health-related quality of life. In addition, observational evidence suggests that maintaining an active lifestyle following cancer diagnosis may contribute to improved overall and cancer-specific survival, particularly among patients with breast and colorectal cancer.

The beneficial effects of exercise are likely mediated through multiple interconnected biological mechanisms, including reductions in chronic inflammation, improvements in insulin sensitivity and metabolic regulation, enhancement of immune surveillance, and favourable modulation of the tumour microenvironment. These complex physiological adaptations support the concept that exercise should be regarded not only as a lifestyle recommendation but also as an important adjunctive component of contemporary cancer care.

Despite the growing body of evidence, important knowledge gaps remain. Considerable heterogeneity among published studies regarding exercise modality, intensity, duration, supervision, and outcome assessment limits direct comparison of findings and highlights the need for greater methodological standardization. Furthermore, additional high-quality randomized controlled trials with long-term follow-up are required to clarify the effects of exercise on disease recurrence, cancer-specific survival, and treatment response, particularly in patients with advanced disease and those receiving novel targeted therapies or immunotherapy.

Future research should focus on developing individualized exercise prescriptions tailored to tumour type, treatment modality, disease stage, and patient characteristics. The integration of precision medicine, digital health technologies, wearable devices, and biomarker-guided exercise interventions may further optimize supportive cancer care and improve long-term clinical outcomes.

From a clinical perspective, the current evidence strongly supports incorporating structured physical activity into routine oncology practice. Collaboration among oncologists, physiotherapists, exercise physiologists, rehabilitation specialists, nurses, and primary care physicians is essential to ensure that exercise is safely and effectively implemented throughout the cancer continuum. As the number of cancer survivors continues to increase worldwide, promoting physical activity should become an integral element of multidisciplinary cancer care aimed at improving both survival and quality of life.

In conclusion, physical activity represents a safe, feasible, and evidence-based supportive intervention with substantial benefits for patients with breast, colorectal, and prostate cancer. Although further research is required to refine optimal exercise prescriptions and clarify the underlying biological mechanisms, current evidence justifies the routine recommendation of individualized exercise programmes as part of standard oncological care.

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